

SHAHID SYED ALI

MECHATRONICS ENGINEERING STUDENT

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SUMMARY OF QUALIFICATIONS

- Third-year BEng + MEng (Hons) Mechatronics Engineering student at the University of Nottingham with a 75% First Class Honours result in Second Year.
- Hands-on robotics experience spanning autonomous mobile robots, ROS/Autoware, LiDAR-based perception, mapping, sensor integration, embedded control and robotic arms.
- Practical experience with TARS autonomous delivery robots and EMERGE autonomous buggies, including testing, troubleshooting, autonomous navigation and charging/parking alignment.
- Strong engineering foundation across CAD, electronics, microcontrollers, manufacturing and control systems, with experience taking systems from design and simulation through fabrication and testing.

PROFESSIONAL EXPERIENCE

Robotics Engineering Intern

HelloWorld Robotics Sdn. Bhd.

Jun 2026 – Sep 2026

Kuala Lumpur, Malaysia

- Worked hands-on with TARS autonomous delivery robots and EMERGE autonomous buggies, supporting robot bring-up, operation, testing and troubleshooting.
- Worked with ROS/Autoware workflows, Ubuntu/SSH tools and Lanelet maps; created and tested a Level 2 company-building map for autonomous navigation.
- Tested five EMERGE buggies and tuned autonomous parking goal coordinates to improve charging-receiver and ground-transmitter alignment, enabling autonomous driving without manual takeover.
- Supported TARS solar-charging integration through wiring, BMS and software testing, and assisted robot demonstrations and technical visits.

SELECTED ENGINEERING PROJECTS

Industrial Conveyor Test Rig + Robot Arm Pick & Place

University of Nottingham

Sep 2025 – Apr 2026

Semenyih, Malaysia

- Implemented an Arduino Mega controller for an ICT3/Bytronic peg-ring verification rig, interfacing 24 V industrial devices with 5 V logic through opto-isolated inputs and relay-driven outputs.
- Integrated a Raspberry Pi and robot arm for pick-and-place using forward/inverse kinematics and motion sequencing; designed in SolidWorks and supported simulation and 3D-printed fabrication.

Centrifugal Clutch Design & Manufacture

University of Nottingham

Sep 2025 – Apr 2026

Semenyih, Malaysia

- Modelled parts and full assembly in SolidWorks; produced BS 8888 manufacturing drawings, BOM/cutting list and process sheets.
- Verified centrifugal/spring forces, pressure and torque against a $\geq 0.3 \text{ N}\cdot\text{m}$ at 2000 rpm requirement and supported CNC fabrication, fitting, assembly and testing.

Advanced Line-Following Robot Car

University of Nottingham

Sep 2024 – Dec 2024

Semenyih, Malaysia

- Built a 4WD Arduino robot with custom IR sensing, ultrasonic obstacle detection, rotary-encoder distance measurement and MPU6050 orientation/ramp sensing.
- Developed embedded control and sensor-integration code, added LCD live data and HC-05 Bluetooth mobile operation.

EDUCATION

BEng + MEng (Hons), Mechatronics Engineering — University of Nottingham, Malaysia Semenyih, Malaysia Sep 2024 – May 2028

- Third Year | Current
- Second Year — 75% | First Class Honours
- First Year — 73% | First Class Honours

Higher Secondary — Al Yasmin International School, Riyadh, Saudi Arabia 2024 • 90.2% (CBSE-India)

TECHNICAL SKILLS

Robotics & Software: ROS/Autoware, Python, C, NumPy, Ubuntu/Linux, embedded control, autonomous navigation

Perception & Mapping: LiDAR, sensor fusion, 2D mapping, Lanelet maps, rotary encoders, ultrasonic/IMU sensing

Engineering & Hardware: SolidWorks, MATLAB, LTspice, Arduino, Raspberry Pi, robot-arm kinematics, ladder logic, CAD, prototyping, CNC/workshop fabrication